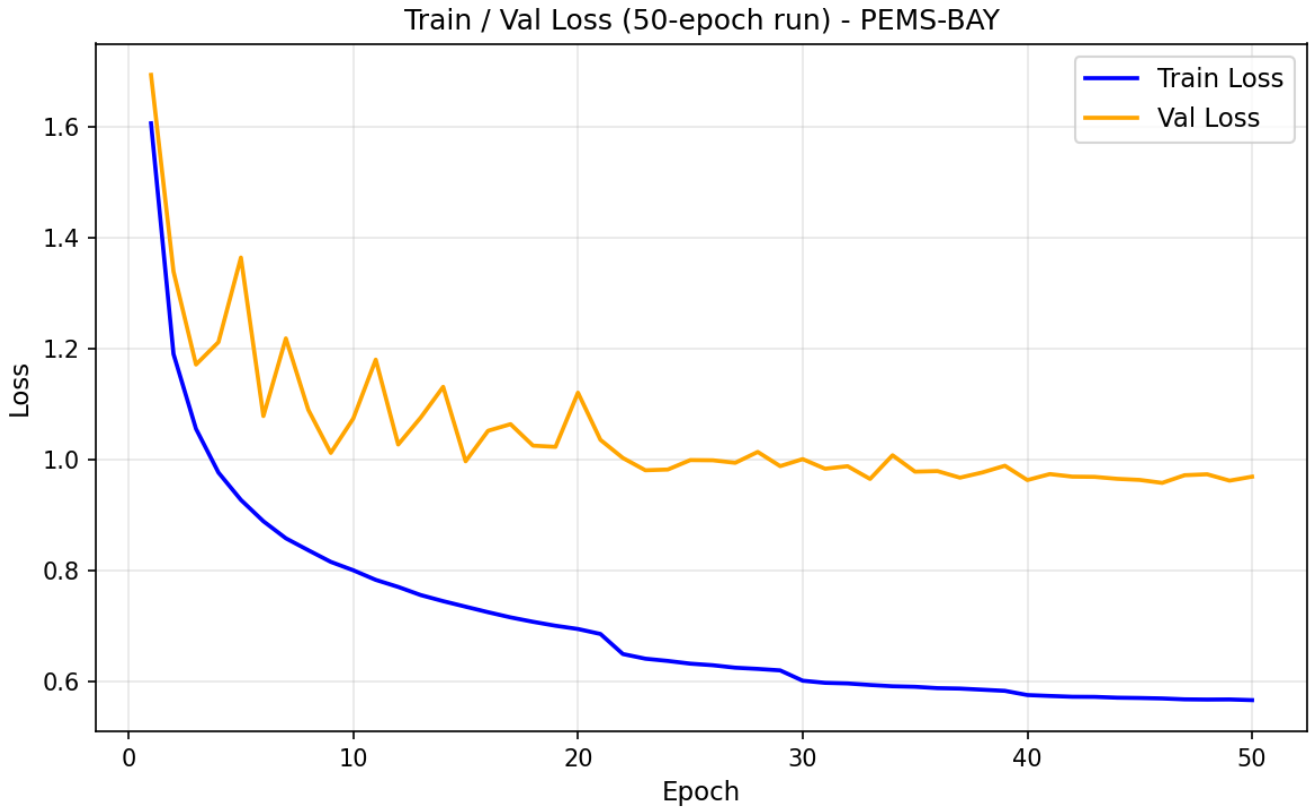
Model and training configuration:

Component	Specification
Model Architecture	
Dataset	PEMS-BAY
Number of Sensors	325
Input Window	12 time steps (60 minutes, 5-min intervals)
Prediction Horizon	12 time steps (60 minutes ahead)
Spatio-Temporal Blocks	4 stacked graph/spatio-temporal layers
Hidden Feature Dimension	64
Diffusion Steps	2-hop propagation
Graph Attention Heads	4
Temporal Convolution	Kernel size 2; dilations {1, 2, 4}
BiLSTM Hidden Size	64
Dropout Rate	0.15
Total Parameters	≈270K (270,340)
Training Setup	
Optimizer	AdamW (learning rate = 10^{-3}, weight decay = 10^{-5})
Scheduler	ReduceLROnPlateau (factor = 0.5, patience = 5)
Epochs / Batch Size	50 / 8
Loss Function	Uncertainty-aware loss: MSE + NLL
Gradient Clipping	max_norm = 1.0
Early Stopping	Patience = 15 epochs
Uncertainty Estimation	
MC Dropout	Enabled at inference
MC Samples	10 stochastic forward passes
Aleatoric Uncertainty	Learned through the variance prediction head
Epistemic Uncertainty	Estimated from variance across MC forward passes

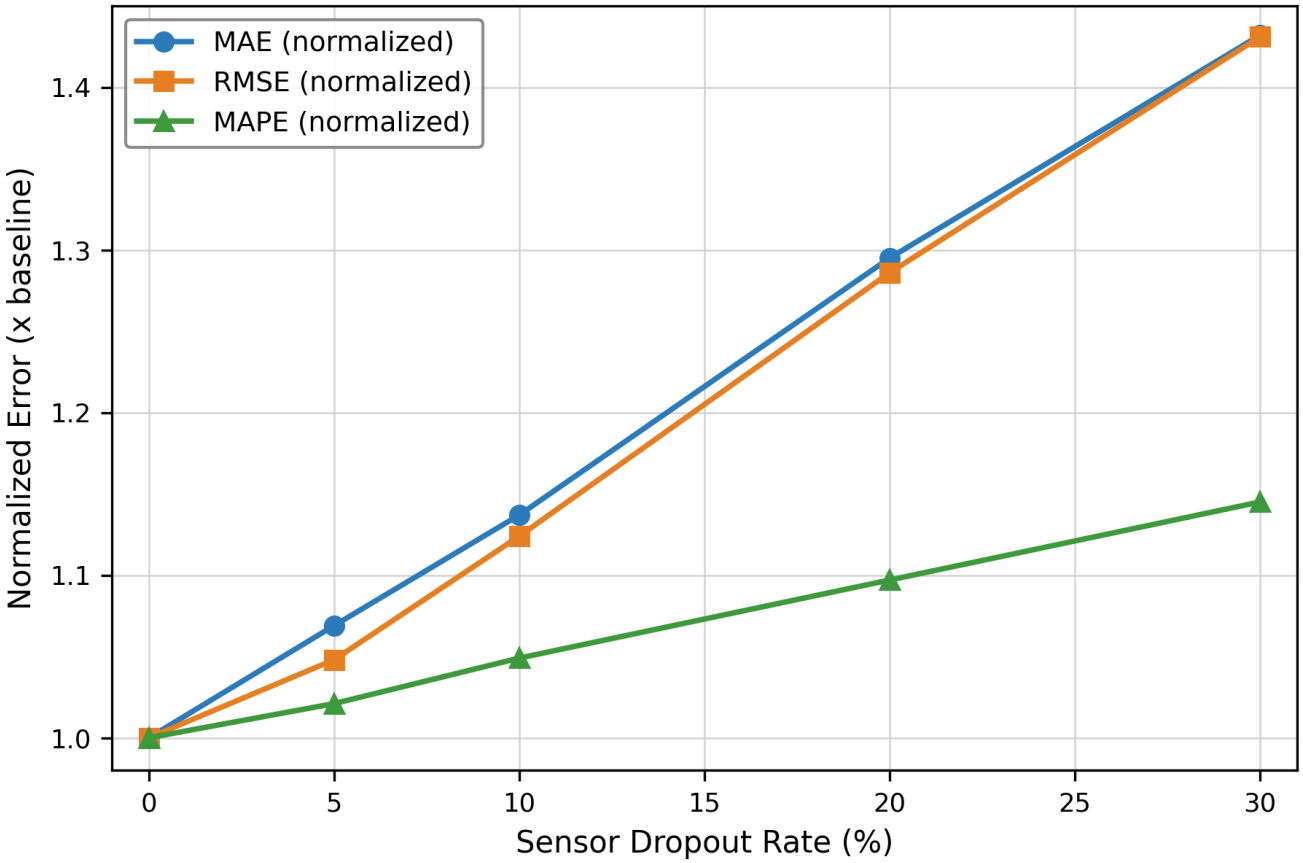
Performance analysis on PEMS-BAY test set:

Metric	Value
MAE	0.4392
RMSE	1.0327
R2 Score	0.8385
Pearson Correlation	0.9158
Mean Aleatoric Uncertainty	0.5508
Mean Epistemic Uncertainty	0.2560
Mean Total Uncertainty	0.8068

Training and validation loss curves over 50epochs, demonstrating smooth convergence and stable gen-eralization behavior on the PEMS-BAY dataset.



Robustness of UA-GNN under sensor dropout on the PEMS-BAY dataset



Correlation matrix of a sensor subset, motivating graph-based spatial modeling.

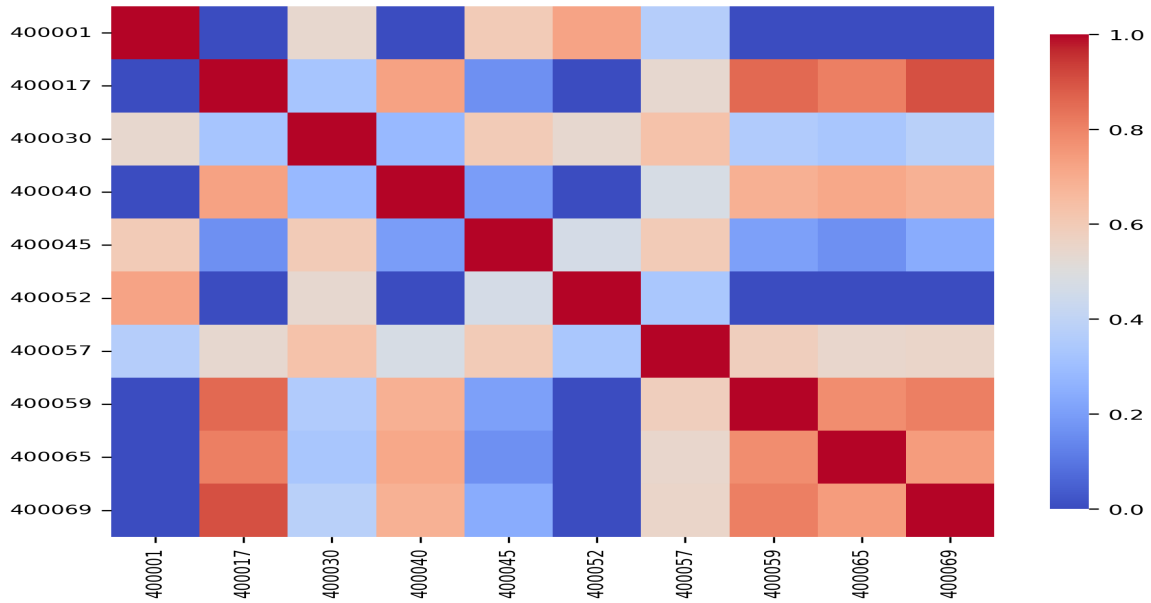


Figure 6: Correlation matrix of a sensor subset, motivating graph-based spatial modeling.

Representative 24-hour traffic speed profile with predictive uncertainty band. Wider intervals correspond to peak-hour variability, while narrower bands indicate stable traffic conditions.

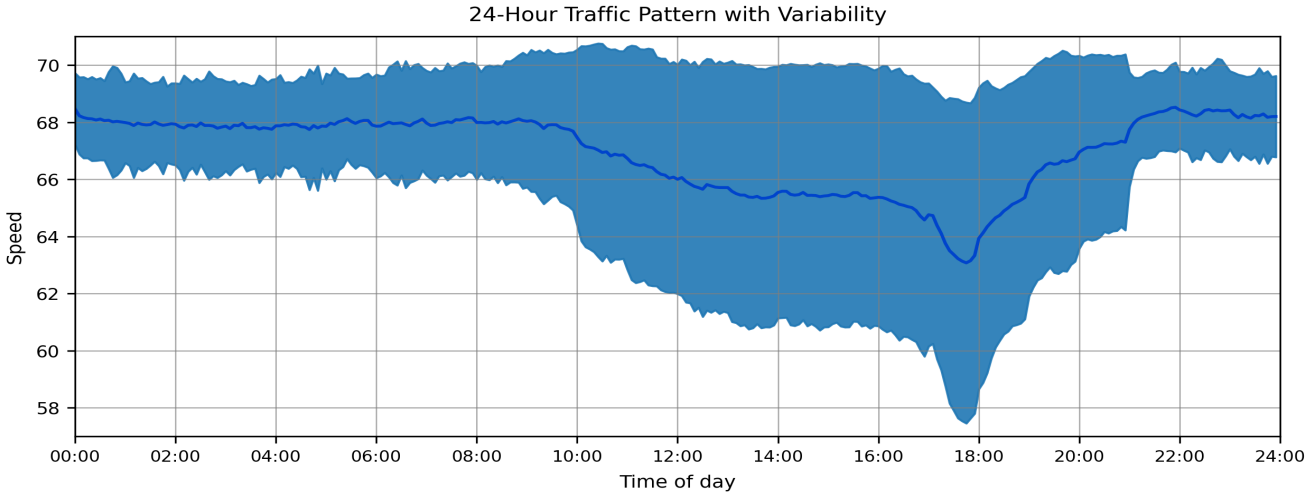


Figure 5: Representative 24-hour traffic speed profile with predictive uncertainty band. Wider intervals correspond to peak-hour variability, while narrower bands indicate stable traffic conditions.

Ablation Study: Component contribution over-all Performance on PEMS-BAY Dataset:

Model Variant	MAE	RMSE	MAE Degradation (%)
Full Model (all components)	0.4392	1.0327	Baseline
w/o Diffusion Conv (GCN only)	23.6171	35.7619	+5277.30%
w/o BiLSTM (Temporal Conv only)	1.5148	2.3970	+244.90%
w/o Graph Attention	1.1911	1.8554	+171.20%
w/o MC Dropout (Deterministic)	0.4392	1.0327	+0.00%

Computational Efficiency Comparison on PEMS-BAY:

Model	Inference Time (ms)	Memory (MB)	Parameters (K)	Notes / Source
Proposed Model	59.3 ± 0.7	358 (train buffer) / 99.8 (batch=32) / 95.4 ± 1.3 (batch=8 eval)	270.3 (core) / 303.8 (w/ embeddings)	Empirically profiled on GPU (exact weights: 270,340)
DCRNN	174.0 ± 3.8	1,417	1,200.0	325 graph node scaling from published baseline
Graph WaveNet	96.2 ± 2.4	803	700.0	325 graph node scaling from published baseline
Bayesian LSTM	125.1 ± 3.1	1,057	900.0	325 node scaling from published baseline

UA-GNN FINAL EVALUATION SUMMARY (PEMS-BAY DATASET)

1. OVERALL METRICS (Standardized / Normalized Latent Space):

- MAE: 0.439200
- RMSE: 1.032700
- R^2 Score: 0.838500
- Pearson Correlation: 0.915800
- Mean Aleatoric Uncertainty: 0.550800
- Mean Epistemic Uncertainty: 0.256000
- Mean Total Uncertainty: 0.806800

2. PHYSICAL SPEED METRICS (Miles Per Hour, mph):

- Overall MAE: 1.6700 mph
- Overall RMSE: 3.7200 mph
- Overall MAPE: 3.85 %

3. HORIZON-SPECIFIC METRICS (Physical mph):

- 15 min (step 3): MAE = 1.3000 mph, RMSE = 2.7400 mph
- 30 min (step 6): MAE = 1.6400 mph, RMSE = 3.7000 mph
- 45 min (step 9): MAE = 1.8800 mph, RMSE = 4.2100 mph
- 60 min (step 12): MAE = 2.0700 mph, RMSE = 4.7400 mph

4. CALIBRATION:

- Expected Calibration Error (ECE): 0.0869